

Operations Research

Mock Exam

— *Do not turn this page before the official start of the exam!* —

First Name, Surname: _____

Student ID: _____

Examiner: Dr. Barbara Franci

Date/time: Wednesday 15th January 2025, 13.00h-15.00h (exam time: 2 hours)

Format: Closed book exam

Allowed aids: Pens, simple (non-programmable) calculator.

Instructions to students:

- The exam consists of 6 questions on 12 pages (excluding the 1 cover page(s)).
- Fill in your name and student ID number on each page, including the cover page.
- Ensure that you properly motivate your answers.
- Write in a readable way. Answers that cannot be read easily cannot be graded and may therefore lower your grade.
- You are not allowed to have a communication device within your reach, nor to wear or use a watch.
- If you think a question is ambiguous, or even erroneous, and you cannot ask during the exam to clarify this, explain this in detail in your answer.
- If you have not registered for the exam, your answers will not be graded, and thus handled as invalid.
- **Good luck!**

The following table will be filled by the examiner(s):

Question:	1	2	3	4	5	6	Total
Points:	15	15	10	20	25	15	100
Score:							

Question 1 (15 points)

Queuing Processes.

For an oral exam, students arrive according to a Poisson distribution with an average of 6 students per hour. There are 3 examiners available for the examination. To each examiner, it takes on average 20 minutes to evaluate a student. If all the examiners are busy, the arriving new student have to wait for their turn. Since the room is small, if there are more than two students waiting, the new arriving students has to come back later. Therefore, there are at most 5 students in the system: 3 taking the exam and 2 waiting.

1. What is the expected number of students in the system at any point in time?
2. Suppose the examiners are ready to start the exams. There are no students waiting yet. How long does it take before a student arrives and has to wait to take the exam? Explain how you build the system of equations to compute such time. You do not need to do the computations.

A large grid of graph paper, consisting of 20 columns and 30 rows of small squares, intended for the student to show their work and calculations.



Question 2 (15 points)

Markov Chains.

We examine a discrete Markov chain with state space $\{1, 2, 3, 4, 5\}$ and transition matrix P as follows

$$P = \begin{pmatrix} P_{11} & P_{12} & P_{13} & P_{14} & P_{15} \\ P_{21} & P_{22} & P_{23} & P_{24} & P_{25} \\ P_{31} & P_{32} & P_{33} & P_{34} & P_{35} \\ P_{41} & P_{42} & P_{43} & P_{44} & P_{45} \\ P_{51} & P_{52} & P_{53} & P_{54} & P_{55} \end{pmatrix} = \begin{pmatrix} 0.6 & 0.4 & 0 & 0 & 0 \\ 0.3 & 0.2 & 0.1 & 0.4 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0.8 & 0.2 \\ 0 & 0 & 0 & 0.9 & 0.1 \end{pmatrix}$$

1. Calculate, for initial state 1, the probability of ever arriving at state 3, and also calculate the probability of ever arriving at state 5.
2. Define transient, recurrent and absorbing states. For each of these terms, state whether or not the provided Markov Chain contains such a state. If it does, give an example of such a state.
3. For the process starting at state 1, calculate for each transient state s , the expected number of stages that the process will be in that particular state s .





Question 3 (10 points)

Game Theory: Two-Player Zero-Sum Games.

Use the graphical method to determine the value and for each player an optimal (mixed) strategy for the matrix game A given by:

$$A = \begin{bmatrix} 6 & 0 \\ -1 & 2 \\ 4 & 1 \end{bmatrix}$$

(As usual player 1 plays rows, player 2 plays columns; player 1 wants to maximize the payoff, while player 2 wants to minimize the payoff to player 1.)





The simplex method for linear programming.

$$\text{maximize } Z = 2x_1 + 3x_2 \quad \text{subject to}$$

$$x_1 + x_2 \leq 12$$

$$x_1 \geq 0, x_2 \geq 0.$$

- Use the tabular form of the simplex method to solve linear program. **Explain** how you choose your *pivots*. Give your solution as $x_1 = \dots, x_2 = \dots, Z = \dots$
- Use the graphical method to find the optimal solution. **Explain** how you do it. Give your solution as $x_1 = \dots, x_2 = \dots, Z = \dots$



Question 5 (25 points)

Transportation problems. [parts a. and b. are unrelated]

- a. In the Transportation table below there are 3 supply points (with supplies 12, 12 and 6) and 3 demand points (with demands 6, 15, 9). The table shows the cost of shipping one unit from each supply point to each demand point. We wish to minimize the shipping costs.

		Destination			
		1	2	3	Supply
Source	1	3	10	2	12
	2	5	4	9	12
	3	5	10	10	6
Demand		6	15	9	

1. Use the *North-west Corner Rule* to identify an initial Basic Feasible Solution (BFS). **State explicitly** the cost of this initial BFS.
 2. Starting from the BFS identified in 1., apply two iterations of the Transportation Simplex method. At each iteration **state explicitly** the current value of the objective function and the values of the basic variables. **Explain** whether, after two iterations, we have reached optimality.
- b. There is a single cookie shop, supplied by a single cookie factory, and both are part of the same business, so the goal is to supply the shop at minimum total cost. Today the cookie shop needs exactly 6kg of cookies, tomorrow it needs exactly 7kg and the day after tomorrow it needs exactly 8kg. Today the cookie factory can produce up to 12kg of cookies at 20 euro per kilogram. Tomorrow the cookie factory can produce up to 4kg of cookies at 40 euro per kilogram. The day after tomorrow it can produce up to 8kg of cookies, at 45 euro per kilogram. The cookies produced by the factory do not have to be shipped to the shop on the same day, but in that case there is a storage cost of 3 euro per kilogram per night. So, for example, 2 kilograms of cookies produced today but shipped only tomorrow will have 6 euro storage costs. However, cookies cannot spend more than 1 night in storage, to prevent them rotting. Note also that a cookie factory does not have to use all its production capacity on a specific day, if it does not want to.
- Explain how you can model this cost-minimization problem as a Transportation Problem. You do not need to solve the problem.



Question 6 (15 points)
Flipping coins.

Consider repeatedly flipping a fair coin. The results of these flips, H's and T's, are listed in a *sequence*, one by one. A finite string of consecutive flips is called a *code*.

- a. Calculate the expected number of flips until the code HHTHH occurs for the first time.
- b. We now are interested in the probability q that the code HTTH appears sooner than THTT in the sequence of flips. Model the problem by a *transition diagram* (graph) and write a system of equations (built from your diagram) using the variables p_x for the probability of reaching HTTH sooner than THTT when starting at state x . Clearly indicate which p_x corresponds to the probability q . You do not need to solve the system.



